Python – One-Line Definitions

Topic	One-line Definition
Python	A high-level, easy-to-learn programming language used to build applications, websites, automation, AI, and data analysis.
Syntax	The set of rules that define how Python code must be written.
Print Statement	print() is used to display output on the screen.
Comments	Comments are notes in the code that Python ignores, usually written using #.
Indentation	Indentation means spaces at the beginning of a line used to define blocks of code in Python.
Case Sensitive	Python treats uppercase and lowercase letters as different, such as Name and name.
Variable	A variable is a name used to store a value in memory.
Python Virtual Environment	A virtual environment is an isolated Python setup used to manage project-specific packages and dependencies.
.py and .ipynb	.py is a Python script file, while .ipynb is a Jupyter Notebook file containing code, output, and explanations.
Arithmetic Operators	Operators such as +, -, *, /, %, and ** are used for mathematical calculations.
Comparison Operators	Operators such as ==, !=, >, <, >=, and <= compare values and return True or False.
Logical Operators	and, or, and not are used to combine or modify conditions.
Assignment Operators	Operators such as =, +=, -=, and *= are used to assign or update values in variables.
Data Types	Data types define the kind of value stored in a variable, such as int, str, boolean, and float.
int	int represents whole numbers, such as 10, 25, or -5.
str	str represents text or a sequence of characters, such as Hello.
Boolean	bool represents one of two values: True or False.
float	float represents numbers containing decimal values, such as 10.5.
List	A list is an ordered and changeable collection of values, written using [].
Tuple	A tuple is an ordered and unchangeable collection of values, written using ().
Set	A set is an unordered collection of unique values, written using {}.
Dictionary	A dictionary stores data as key-value pairs, written using {key: value}.
Control Flow	Control flow determines the order in which Python statements are executed.
Nested If	A nested if is an if statement placed inside another if statement.
Loop	A loop repeatedly executes a block of code until a condition or sequence is completed.
Function	A function is a reusable block of code designed to perform a specific task.
Module	A module is a Python file containing reusable code such as functions, classes, and variables.
Exception Handling	Exception handling manages runtime errors using try, except, else, and finally.
File Handling	File handling allows Python to create, read, write, and modify files.